

1.1.3 Analog and Digital Signals

اینالگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

CS XI

Chapter 1: Computer Systems

Lecture 02 of 12

★ Golden

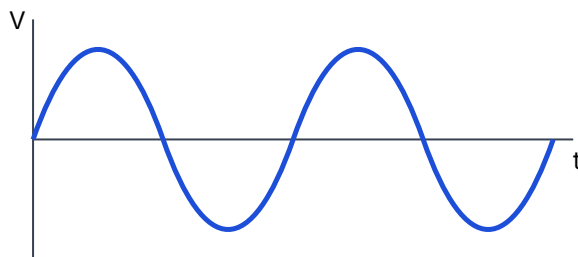
1.1.3 Analog and Digital Signals ★ Golden

LEARN IT

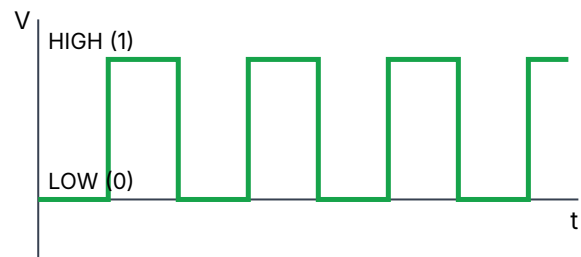
A **signal** is a changing voltage (or other quantity) that carries information.

Analog signals are continuous signals that change smoothly over time, like a **sine wave**. Examples: the human voice, a traditional mercury thermometer.

Digital signals have only two levels: **HIGH (1)** and **LOW (0)**. They jump instantly between these values, which creates a **square wave**. Example: computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.

WORKED EXAMPLE

Why is digital more resistant to noise?

Step 1: Suppose a wire carries 5 V for HIGH and 0 V for LOW.

Step 2: Noise adds a small error, so 5 V arrives as 4.6 V and 0 V arrives as 0.3 V.

Step 3: The receiver only asks "Is it closer to HIGH or LOW?" 4.6 V is still read as 1 and 0.3 V is still read as 0. The data is perfect.

Step 4: In an analog signal, every exact value carries meaning, so 4.6 V instead of 5 V really changes the sound or reading. The noise cannot be removed.

REMEMBER IT

Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

COMMON MISTAKE

Do not draw a digital signal with sloping sides. A digital signal jumps *instantly*, so its edges are straight vertical lines. Also do not confuse "digital" with "electronic": an analog signal can also be electronic.

اردو میں سمجھیں

اینالوگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔ ڈیجیٹل سگنل کی صرف دو سطحیں ہوتی ہیں، HIGH (1) اور LOW (0)، اور یہ اچانک ایک سے دوسری پر جاتا ہے، اس لیے مربع لہر (Square Wave) بنتی ہے۔ ڈیجیٹل سگنل شور سے کم متاثر ہوتا ہے، اس لیے زیادہ قابلِ بھروسہ ہے۔

EXAM TIP

Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave shape, reliability, example). Always label your diagrams: axes V and t, and HIGH(1)/LOW(0) on the square wave.

CHECK YOURSELF

1. What shape of wave does a digital signal make?
2. Which type of signal is more affected by noise?
3. Give one example of an analog signal.

Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An analog signal:

- a. Changes smoothly, like a sine wave
- b. Jumps only 0/1
- c. Is a truth table
- d. Is TCP

MCQ 2. A digital signal looks like:

- a. A square wave of HIGH and LOW
- b. A smooth sine
- c. A pie chart
- d. A stack

MCQ 3. Noise harms analog more because:

- a. Many in-between levels can be bent
- b. Bits cannot exist
- c. OSI has 7 layers
- d. Python is analog

MCQ 4. Human voice on a microphone is first:

- a. Analog
- b. A Boolean 1 only
- c. A K-map
- d. SQL

MCQ 5. HIGH in digital means:

- a. 1 / ON / TRUE
- b. 0 / OFF
- c. A continuous ramp
- d. Gray code

MCQ 6. Copying analog tape many times:

- a. Loses quality
- b. Stays perfect
- c. Creates bits
- d. Builds NAND

Extra short questions

1. Define analog signal and digital signal.
2. Give one example of each.
3. Why is digital more reliable?
4. What does a computer do with analog sound?

Extra long question

1. Draw analogue (sine) and digital (square) waves. Tabulate nature and reliability. This is a ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Analog: continuous, smooth change. Digital: only two levels, HIGH(1) and LOW(0), a square wave.
2. Analog: traditional thermometer / voice. Digital: computer data / logic pulse.
3. A bent 0 or 1 can be restored to a clean 0 or 1; analog distortion stays in the wave.
4. Sample and quantise it into bits, process the bits, then convert back to analog for a speaker.

Long

1. Sine labelled V against t; square labelled HIGH/LOW. Table: analog = continuous, low reliability; digital = discrete bits, high reliability. Close with the noise argument.

Next lecture: 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables